MODULE 26 · DOMAIN 4 - IMPLEMENT TEXT ANALYSIS SOLUTIONS

Sentiment, sensitive content and domain customisation

Sentiment and tone with models, harmful text, PII detection and redaction policies, compliance summarisation, and when to fine-tune.

DOMAIN 4 - Text analysis WEIGHT 10-15% MODULE 26 of 39

admin@sthithapragna.com

Detection tells you what's in the text. **Customisation** changes what the model gives back.

Which service detects what?

TO DETECT	USE	STATUS
Sentiment and tone	A model with a schema	Recommended for new work
Sentiment, opinions	Azure Language sentiment and opinion mining	Legacy, retiring by 2029
Harmful text	Content Safety text analysis, guardrails	Current
Prompt attacks	Prompt Shields	Current
Personal data	Azure Language PII detection	Core, text PII GA
Health information	Text analytics for health	Core

ON THE EXAM

Each row answers a different question. A PII detector doesn't find hate speech, and a harm filter doesn't find phone numbers.

Detecting sentiment with a schema

- Use an **enum** for the label: positive, negative, neutral or mixed.
- Add **aspects**: an array of what the writer liked or disliked, each with its own label.
- Ask for an **evidence quote** behind each judgement.
- **Calibrate** against a labelled sample, and measure agreement with an evaluator.
- Keep instructions consistent: label drift between prompt versions breaks trend reports.

> ON THE EXAM
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> 'Food great, service slow' is mixed overall, but positive for food and negative for service. Aspects capture that.

Azure Language sentiment and opinion mining

- Labels for the **document** and each **sentence**: positive, neutral or negative, with **confidence scores**.
- A document can also be **mixed**, when its sentences disagree.
- **Opinion mining** links opinions to **targets**: 'slow' is about 'service'.
- Up to 10 documents per request, with containers available for on-premises use.
- **Legacy:** supported for existing implementations, and retiring by March 2029.

> ON THE EXAM
>
> Still valid in an existing app, and still examinable. For a new build, a model with aspects is Microsoft's direction.

How someone feels, or how they say it

SENTIMENT

Polarity

- Positive, negative, neutral, mixed
- About the topic being discussed
- Classic feedback analytics

TONE

Manner

- Frustrated, urgent, formal, sarcastic
- About how the message is written
- Routing, escalation, brand voice

THE TIEBREAKER

Azure Language has no tone feature. Tone, urgency and sarcasm come from a model with defined labels.

Detecting harmful text

- **Content Safety text analysis**: hate, sexual, violence and self-harm, with severity levels.
- Model deployments are guarded automatically. **User-generated text** you store or publish needs your own call.
- **Blocklists** catch exact terms; **custom categories** catch patterns you define.
- **Prompt Shields** detect attacks in prompts and documents; **protected material** detects copyrighted text.
- Guardrail thresholds and actions work as module 09 showed.

> ON THE EXAM
>
> A forum post never sent to a model is never seen by a model guardrail. Moderate it with the Content Safety API.

PII detection in Azure Language

- **Text PII** is generally available, from API version 2026-05-01.
- **Conversation PII** handles chats and transcripts; **document PII** redacts PDF, Word and text files.
- Document PII **preserves layout**, and returns a redacted file plus JSON results.
- Set a **confidence threshold**, **exclude values** you know are safe, and add **entity synonyms**.
- Limit which types are detected with **piiCategories**.

> ON THE EXAM
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> Excluding known safe values, like your own support phone number, stops needless redaction.

Four ways to redact

POLICY	OUTPUT	USE IT WHEN
`characterMask`	'John Doe' becomes '********'	Hide values but keep length and offsets
`entityMask`	'John Doe' becomes a Person label	Readers need to know what was removed
`syntheticReplacement`	'John Doe' becomes 'Sam Johnson'	Text must stay natural, like training data
`noMask`	No redacted text returned	You only need the detected entities

ON THE EXAM

One request can set a default policy, then override it for specific entity types.

Synthetic replacement for names and phone numbers

```
POST {endpoint}/language/:analyze-text?api-version=2026-05-01
{"kind": "PiiEntityRecognition",
 "parameters": {
   "modelVersion": "latest",
   "redactionPolicies": [
     {"policyKind": "syntheticReplacement",
      "entityTypes": ["Person", "PhoneNumber"]}]},
 "analysisInput": {"documents": [
   {"id": "1", "language": "en",
    "text": "John Doe received a call from 424-878-9193."}]}}

# redactedText: "Sam Johnson received a call from 401-255-6901."
```

Masks hide. Labels explain. Synthetic values keep text natural.

Redact before data spreads

- Redact **before** text is logged, indexed for search, or sent to a model.
- Traces and search indexes keep copies, so late redaction leaves data behind.
- Use **conversation PII** for transcripts, where spoken language breaks written patterns.
- A **PII guardrail** (preview) can also act at model inputs and outputs, as module 09 showed.
- Redaction reduces risk; it doesn't replace access control on the original data.

> ON THE EXAM
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> Module 16's lesson again: once personal data is in a trace, it's in telemetry.

The customisation ladder

01
Instruct
Clear rules, roles, output contract.

02
Show
Few-shot domain examples.

03
Structure
A strict schema for the output.

04
Ground
Retrieve current policy and terms.

05
Evaluate
Measure before changing more.

06
Fine-tune
Only for behaviour that won't stick.

ON THE EXAM

Climb one rung at a time, and measure at each. Most domain tasks never need the top rung.

A compliance summary pattern

- **Redact** personal data first, so it never reaches the model or the summary.
- **Retrieve** the current regulation or policy, so the summary uses today's rules.
- Require fixed sections in a schema: **obligations, deadlines, risks, and clause citations**.
- **Evaluate** groundedness and completeness, as module 15 showed.
- Keep a **person** accountable for sign-off. The summary supports the decision.

> ON THE EXAM
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> 'Summarise this for compliance' is vague. Required sections with citations make the output reviewable.

Microsoft's reasons to fine-tune

- **Prompts have grown long** with few-shot examples, raising token cost and latency.
- You need a **consistent style or tone**, such as a brand voice.
- Outputs must follow **specific formats or schemas** that prompting doesn't hold reliably.
- The model must use **many tools**, and listing them all costs tokens and accuracy.
- **Not** for facts that change. Changing knowledge belongs in retrieval, as module 03 showed.

> THE TIEBREAKER
>
> Fine-tuning teaches behaviour. Retrieval supplies knowledge. Monthly-changing rules plus fine-tuning is the classic wrong answer.

SFT, DPO or RFT?

METHOD	TRAINS ON	BEST FOR
Supervised fine-tuning (SFT)	Labelled input and output pairs	Most scenarios, including task specialisation
Direct preference optimisation (DPO)	Preferred and non-preferred responses	Aligning quality and style with preferences
Reinforcement fine-tuning (RFT)	Reward signals from graders	Complex behaviours, often reasoning tasks

ON THE EXAM

Methods stack: SFT first for the task, then DPO to align responses with preferences.

Doing fine-tuning properly

- Training data is **JSONL**, in the **Chat Completions conversation format**.
- **Evaluate the base model first**, so improvement can be measured.
- The **Standard** training tier keeps training in your resource's region, for data residency.
- **Compare checkpoints** rather than deploying the final one blindly.
- With RFT, rewards that **diverge** suggest **reward hacking**: the grader needs work.

> ON THE EXAM
>
> You can fine-tune a model that was already fine-tuned, such as adding DPO on top of an SFT model.

Requirement to approach

REQUIREMENT	APPROACH
Sentiment and urgency for a new feedback app	Model schema with enums, aspects, evidence
Moderate forum posts before publishing	Content Safety text analysis
Transcripts without real names, still readable	PII detection, syntheticReplacement
Reviewers must see what type was removed	PII detection, entityMask
Summaries against rules that change monthly	Retrieval plus a strict schema, not fine-tuning
A brand voice prompting can't hold	Supervised fine-tuning
Correct outputs, but a preferred phrasing	DPO on preference pairs

ON THE EXAM

Detect first, redact before spreading, then climb the customisation ladder only as far as you must.

> A bank wants to build a customer-service chatbot trained on past support chats. The chats contain real customer names and phone numbers.
>
> The training data must read like natural conversation, but contain no real personal details.

Which redaction policy should the team use?

A characterMask, replacing every name and number with asterisks.

B noMask, which returns detected entities without redacted text.

C entityMask, replacing each value with its entity type label.

D syntheticReplacement, swapping in realistic substitutes.

◆ PAUSE AND DECIDE

Which redaction policy should the team use?

A characterMask.

Hides values, but asterisks teach the chatbot to talk in asterisks.

B noMask.

Returns no redacted text at all, so nothing is removed from the chats.

C entityMask.

Type labels are clear for reviewers, but unnatural in training conversations.

D syntheticReplacement.

Realistic substitutes keep chats natural, with no real details left.

THE SIGNAL

'Must read naturally' plus 'no real personal data' is synthetic replacement.

> A utility company is building a new app to triage customer emails. It needs each email's sentiment, and whether the customer sounds frustrated or urgent.
>
> Urgent, frustrated emails must jump the queue.

What should the team use?

A Azure Language sentiment analysis, using negative scores as urgency.

B A model schema with sentiment, tone and urgency, plus evidence.

C Content Safety hate detection, treating anger as a harm category.

D Azure Language key phrase extraction, searching phrases for 'urgent'.

◆ PAUSE AND DECIDE

What should the team use?

A. Language sentiment as urgency.
Negative isn't urgent. A calm complaint and a burst pipe score alike. Legacy too.

B. Model schema with tone and urgency.
Defines exactly the labels triage needs, with evidence to check them.

C. Hate detection for anger.
Frustration isn't hate. The harm classifier won't flag ordinary anger.

D. Key phrases for 'urgent'.
Misses urgency phrased any other way, and it's a legacy feature.

THE SIGNAL

Tone and urgency aren't sentiment. There's no Language tone feature, so define the labels in a schema.

A firm fine-tuned a model with SFT to write compliance summaries. The summaries are accurate and correctly formatted.

Reviewers still consistently prefer one of two phrasing styles, and have collected hundreds of pairs of preferred and rejected summaries.

What should the team do next?

A Run DPO on the preference pairs, on top of the fine-tuned model.

B Run another round of SFT, adding more correct example summaries.

C Run RFT with a grader that scores summaries for factual accuracy.

D Add retrieval over the regulations so the style matches the source.

◆ PAUSE AND DECIDE

What should the team do next?

A DPO on preference pairs.

Built for preferred versus rejected responses, and stacks on an SFT model.

B More SFT examples.

Outputs are already correct. SFT doesn't learn from 'this one, not that one'.

C RFT for accuracy.

Accuracy is already fine. The gap is preference, and the data is pairs.

D Retrieval for style.

Retrieval supplies facts, not preferred phrasing.

THE SIGNAL

Correct outputs, a preferred style, and preferred-versus-rejected pairs: DPO, stacked on SFT.

> A compliance assistant summarises contracts against financial regulations. The regulations are updated most months.
>
> The team proposes fine-tuning the model every month on the latest regulations, so summaries stay current.

What should the team do instead?

A Run supervised fine-tuning each month on the newest regulation text.

B Run DPO each month on preferred and rejected monthly summaries.

C Retrieve current rules, with a strict schema requiring citations.

D Raise temperature so the model adapts to new regulations more readily.

◆ PAUSE AND DECIDE

What should the team do instead?

A. Monthly SFT.
Fine-tuning teaches behaviour, not reliable facts, and retraining monthly is slow and costly.

B. Monthly DPO.
Aligns style preferences; it doesn't supply this month's rules.

C. Retrieval with a strict schema and citations.
Current rules at request time, with every obligation traceable to a clause.

D. Higher temperature.
Adds randomness, not knowledge of new regulations.

THE TIEBREAKER

Knowledge that changes belongs in retrieval. Fine-tune only the stable behaviour, if needed at all.

Detection and customisation in five lines

- **Sentiment and tone:** a model schema for new work. Language sentiment is legacy.
- **Harm:** Content Safety. **Personal data:** PII detection. Different questions.
- **Redaction:** masks hide, labels explain, synthetic values keep text natural.
- **Climb the ladder:** instruct, show, structure, ground, evaluate, then fine-tune.
- **SFT** for tasks, **DPO** for preferences, **RFT** for graded behaviour. Never for changing facts.

> ON THE EXAM
>
> Next, module 27: translation with Azure Translator and language models.